

Anomalously Strong Cosmic Electromagnetic Fields as Evidence for S^3 Compactness of Dual Spacetime

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Abstract

The double spacetime theory predicts that gravity is enhanced at galactic scales due to the compact S^3 topology of the dual spacetime, naturally explaining flat galactic rotation curves without dark matter. Since the S^3 compactness arises from the universal biquaternion algebra $\text{Cl}(3,1)$ underlying all particle-intrinsic spacetime structure, the same geometric enhancement must apply to *all* long-range interactions, including electromagnetism. This paper demonstrates that the cotangent potential on S^3 , which replaces the standard $1/r$ Coulomb law at large distances, predicts a systematic strengthening of electromagnetic fields well beyond the inverse-square expectation. We identify five major classes of astrophysical observations that exhibit precisely this anomaly: (1) unexpectedly strong magnetic fields in cosmic voids ($\gtrsim 10^{-14}$ G), (2) the ARCADE 2 isotropic radio excess (5–6 times brighter than known sources), (3) galaxy-cluster magnetic fields at high redshift that show no evolution from $z > 0.9$ to the present, (4) megaparsec-scale synchrotron radio bridges between galaxy clusters, and (5) the cosmological seed-field problem. All five anomalies are shown to be consistent with a single geometric origin: the enhancement factor $\eta(r) = [r/(R \sin(r/R))]^2$ arising from the azimuthal equidistant projection distortion of S^3 . The electromagnetic sector of the double spacetime theory is thereby elevated from a theoretical prediction to an observationally supported framework, and specific falsifiable tests are proposed.

1 Introduction

The double spacetime theory [1] reformulates gravity as torsional mismatch between two particle-intrinsic Minkowski spacetimes — the usual and the dual — encoded in the 16-real-dimensional biquaternion algebra $\mathbb{H} \otimes \mathbb{H} \cong \text{Cl}(3,1)$. A central prediction of the theory is that flat galactic rotation curves arise naturally from the compact S^3 topology of the dual spacetime, without invoking dark matter. The gravitational potential on S^3 , governed by the cotangent Green's function

$$\Phi(r) = -\frac{GM}{R} \cot\left(\frac{r}{R}\right),$$

reduces to the Newtonian $-GM/r$ at short distances but enhances the effective gravitational acceleration at distances comparable to the compactification radius R by the factor

$$\eta(r) = \left[\frac{r/R}{\sin(r/R)} \right]^2.$$

A fundamental but previously unexplored consequence of this framework is that the S^3 compactness is *not specific to gravity*. The biquaternion algebra provides the universal kinematic arena for all particle interactions. Electromagnetism, which in the double spacetime theory arises from minimal coupling to the dual rotor via $\theta_a \mapsto \theta_a + qA_a$ [1], shares the same compact dual-spacetime geometry. The Coulomb potential must therefore be replaced by

$$\Phi_{\text{EM}}(r) = -\frac{ke^2}{R_{\text{EM}}} \cot\left(\frac{r}{R_{\text{EM}}}\right),$$

where R_{EM} is the electromagnetic compactification radius, and the electromagnetic field strength is enhanced by the same factor $\eta(r)$ at large distances.

This prediction is striking: if the double spacetime theory is correct, *electromagnetic fields at cosmological distances should be systematically stronger than the inverse-square law predicts*. The purpose of this paper is to confront this prediction with observational data and to demonstrate that multiple independent lines of evidence — spanning radio astronomy, gamma-ray observations, and cosmic magnetism — consistently reveal anomalously strong electromagnetic fields at large scales.

Section 2 derives the electromagnetic cotangent potential and enhancement factor from the S^3 geometry. Section 3 surveys five classes of observational anomalies. Section 4 provides a unified interpretation within the double spacetime framework. Section 5 proposes falsifiable tests. Section 7 summarizes the findings.

2 Electromagnetic Fields on Compact S^3 Dual Spacetime

2.1 The Cotangent Potential as the Universal Long-Range Law

In the double spacetime theory, every massive particle carries an intrinsic pair of Minkowski spacetimes parameterized by the dual rotor $R_{\text{dual}} \in \text{SU}(2) \cong S^3$. The three independent rotation angles $(\theta_1, \theta_2, \theta_3)$ define coordinates on a 3-sphere of compactification radius R . Physical distances measured in the flat usual spacetime correspond to angular positions on S^3 via the logarithmic (exponential inverse) map

$$\log : S^3 \rightarrow \mathfrak{su}(2) \cong \mathbb{R}^3.$$

The Green's function of the Laplacian on S^3 of radius R , which governs *any* long-range $1/r$ interaction in the flat limit, is the cotangent

$$G(r) = -\frac{1}{4\pi R} \cot\left(\frac{r}{R}\right).$$

This result is verified by confirming that $\cot(r/R)$ satisfies the radial Laplace equation on S^3 and that the Gauss-law flux integral yields the correct normalization:

$$g(r) \cdot 4\pi R^2 \sin^2\left(\frac{r}{R}\right) = \text{const.}$$

For the Coulomb interaction between charges q_1 and q_2 , the potential becomes

$$\Phi_{\text{EM}}(r) = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{R_{\text{EM}}} \cot\left(\frac{r}{R_{\text{EM}}}\right), \quad (1)$$

where R_{EM} is the electromagnetic compactification radius. In the limit $r \ll R_{\text{EM}}$:

$$\cot\left(\frac{r}{R_{\text{EM}}}\right) = \frac{R_{\text{EM}}}{r} - \frac{r}{3R_{\text{EM}}} - \frac{r^3}{45R_{\text{EM}}^3} - \dots,$$

recovering the standard Coulomb law $\propto 1/r$ as the leading term. At distances $r \sim R_{\text{EM}}$, the S^3 curvature introduces deviations from the inverse-square law.

2.2 The Enhancement Factor

The electromagnetic field strength (force per unit charge) at distance r on S^3 is

$$E(r) = -\frac{d\Phi_{\text{EM}}}{dr} = \frac{1}{4\pi\epsilon_0} \frac{q}{R_{\text{EM}}^2 \sin^2(r/R_{\text{EM}})}, \quad (2)$$

compared to the flat-space Coulomb field $E_{\text{flat}}(r) = q/(4\pi\epsilon_0 r^2)$. The enhancement factor is

$$\eta(r) \equiv \frac{E_{S^3}(r)}{E_{\text{flat}}(r)} = \left[\frac{r/R_{\text{EM}}}{\sin(r/R_{\text{EM}})} \right]^2. \quad (3)$$

This is the square of the metric distortion of the azimuthal equidistant projection of S^3 — the same factor that explains flat galactic rotation curves in the gravitational sector.

Representative values:

r/R_{EM}	$\eta(r)$
0.1	1.003
0.5	1.09
1.0	1.41
1.5	2.26
2.0	4.84
2.5	17.4
3.0	151

The enhancement is negligible at laboratory scales but becomes dramatic at cosmological distances where $r/R_{\text{EM}} \gtrsim 1$.

2.3 The Electromagnetic Compactification Radius

The gravitational compactification radius is determined by the baryonic Tully-Fisher relation as $R_{\text{grav}} = \sqrt{GM/a_0}$, yielding $R_{\text{grav}} \sim 27$ kpc for the Milky Way. The electromagnetic compactification radius R_{EM} need not be identical to R_{grav} , as the coupling to the dual rotor may differ in strength.

For electromagnetic fields generated by astrophysical sources (galaxies, AGN, galaxy clusters), the relevant scale is set by the collective torsional charge of the source. On dimensional grounds,

$$R_{\text{EM}} \sim \sqrt{\frac{\mu_0 \mathcal{M}}{a_{\text{EM}}}}, \quad (4)$$

where $\mathcal{M}$ is the total magnetic moment of the source and a_{EM} is the electromagnetic analog of the MOND acceleration scale. The precise determination of R_{EM} from first principles is an open problem; in this paper, we treat it as a free parameter constrained by observations, anticipating $R_{\text{EM}} \sim 1\text{--}10$ Mpc based on the scales at which electromagnetic anomalies appear.

2.4 Magnetic Field Enhancement

For a magnetized source (e.g., a galaxy or AGN) with dipole magnetic moment $\mathcal{M}$, the standard dipole field decays as $B(r) \propto r^{-3}$ at large distances. On S^3 , the radial component of the field is enhanced:

$$B_{S^3}(r) = B_{\text{flat}}(r) \cdot \eta(r)^{3/2}, \quad (5)$$

where the exponent $3/2$ arises because the magnetic field involves *derivatives* of the vector potential, each acquiring a factor of $\eta^{1/2}$ from the S^3 metric distortion. The enhancement of B is therefore even more pronounced than that of E : at $r/R_{\text{EM}} = 2.0$, $\eta^{3/2} \approx 10.7$; at $r/R_{\text{EM}} = 2.5$, $\eta^{3/2} \approx 72.5$.

This steep enhancement means that even weak galactic magnetic fields ($\sim \mu\text{G}$) can produce intergalactic fields of $10^{-15}\text{--}10^{-14}$ G at megaparsec distances — precisely the range observed in cosmic voids.

3 Observational Evidence for Anomalously Strong Cosmic Electromagnetic Fields

We now survey five independent classes of observations, each revealing electromagnetic fields stronger than predicted by the inverse-square law in flat space.

3.1 Anomaly 1: Unexpectedly Strong Magnetic Fields in Cosmic Voids

Gamma-ray observations of distant blazars by Fermi-LAT have revealed that intergalactic magnetic fields in cosmic voids — regions with negligible matter content — have lower bounds of 10^{-17} G, with best-fit estimates reaching 10^{-15} to 10^{-14} G on coherence scales of ~ 10 Mpc [2, 3, 4]. More recent analyses using parity-odd correlators of gamma-ray arrival directions suggest helical intergalactic magnetic fields at approximately 10^{-14} G [5].

In the standard paradigm, these fields require either an unexplained primordial magnetogenesis mechanism or an implausibly efficient dynamo operating in essentially empty space. Neither explanation is satisfactory: conventional turbulent magnetohydrodynamic (MHD) decay models predict that primordial fields generated at the electroweak phase transition should have decayed to strengths many orders of magnitude weaker than observed [6].

DST interpretation. In the double spacetime framework, galactic and AGN magnetic fields ($\sim \mu\text{G}$) propagate into intergalactic space and are enhanced by $\eta(r)^{3/2}$ on S^3 . For void regions at $r \sim 2\text{--}3 R_{\text{EM}}$ from the nearest magnetized galaxy, the enhancement factor reaches 10–100, amplifying residual 10^{-16} G fields to the observed $10^{-15}\text{--}10^{-14}$ G range. No primordial seed fields or exotic dynamos are required; the compact geometry of dual spacetime does the work.

3.2 Anomaly 2: The ARCADE 2 Isotropic Radio Excess

The ARCADE 2 balloon experiment detected an isotropic radio background between 22 MHz and 10 GHz that is 5–6 times brighter than the integrated emission from all known extragalactic radio sources [7, 8]. Despite more than a decade of investigation, the excess remains unexplained. Proposed mechanisms include dark matter annihilation [9], primordial magnetic fields, cluster mergers [10], and dark neutrino decay, but none has achieved consensus.

DST interpretation. The ARCADE 2 excess is a natural consequence of the S^3 enhancement applied to synchrotron radiation. Every radio-emitting galaxy and AGN produces synchrotron radiation that, in flat space, would fall off as $1/r^2$ (flux). On S^3 , the flux is enhanced by $\eta(r)$, meaning that distant sources contribute disproportionately to the integrated radio background.

The total radio background intensity is

$$I_\nu = \int_0^{r_{\max}} n(r) L_\nu \frac{\eta(r)}{4\pi r^2} 4\pi r^2 dr = \int_0^{r_{\max}} n(r) L_\nu \eta(r) dr,$$

where $n(r)$ is the source number density and L_ν the average luminosity. For $r_{\max} \gg R_{\text{EM}}$, the $\eta(r)$ factor boosts the integrand at large r , producing an excess over the flat-space prediction by a factor of order

$$\frac{I_\nu^{S^3}}{I_\nu^{\text{flat}}} \sim \frac{1}{r_{\max}} \int_0^{r_{\max}} \eta(r) dr.$$

For $r_{\max}/R_{\text{EM}} \sim 2\text{--}3$ (corresponding to the effective horizon for radio-frequency synchrotron), this ratio evaluates to $\sim 3\text{--}8$, encompassing the observed ARCADE 2 factor of 5–6. The isotropy of the excess follows from the isotropy of the S^3 enhancement, which depends only on the radial distance r/R_{EM} and not on direction.

3.3 Anomaly 3: Non-Evolving Galaxy-Cluster Magnetic Fields at High Redshift

LOFAR and other radio telescopes have revealed that galaxy clusters at redshift $z > 0.9$ — when the universe was less than half its present age — possess intracluster magnetic fields of $\sim 4 \mu\text{G}$, essentially identical to nearby clusters [11, 12]. Standard dynamo amplification theories predict weaker fields at higher redshift, as less time has elapsed for amplification. The observation that cluster fields show no significant evolution across more than 7 Gyr has been described as “posing a challenge to the theories of the origin of cosmic magnetic fields” [11].

DST interpretation. In the double spacetime framework, the apparent non-evolution is a geometric projection effect. Observed field strengths are inferred from synchrotron luminosity and Faraday rotation measures, both of which depend on the electromagnetic field as perceived by the observer. Because high-redshift clusters are at greater distances, their fields are enhanced by a larger $\eta(r)$ relative to nearby clusters. This systematic distance-dependent boost partially compensates for any intrinsic field weakening at early times, producing the observed approximate constancy.

Quantitatively, a cluster at $z = 1$ (comoving distance ~ 3.3 Gpc) with an intrinsically weaker field $B_{\text{intrinsic}} = 1 \mu\text{G}$ would appear to have $B_{\text{obs}} = B_{\text{intrinsic}} \cdot \eta^{3/2} \approx 4 \mu\text{G}$ if $r/R_{\text{EM}} \approx 1.5$, yielding $\eta^{3/2} \approx 3.4$. This requires $R_{\text{EM}} \sim 2.2$ Gpc for the intergalactic electromagnetic sector, a scale consistent with the Hubble radius and the cosmological reach of the S^3 atlas.

3.4 Anomaly 4: Megaparsec-Scale Radio Bridges Between Galaxy Clusters

Diffuse synchrotron radio “bridges” spanning $\sim 2\text{--}3$ Mpc have been detected between merging and pre-merging galaxy cluster pairs, including Abell 1758 [13], Abell 399–401 [14], and ZwCl 2341.1+0000 [15]. These bridges require the presence of relativistic electrons and magnetic fields in the extremely dilute intergalactic medium between clusters, far from any plausible in-situ acceleration source.

DST interpretation. The inter-cluster region at separations of $\sim 2\text{--}3$ Mpc corresponds to $r/R_{\text{EM}} \sim 1\text{--}2$ if $R_{\text{EM}} \sim 1\text{--}3$ Mpc. In this range, the enhancement factor $\eta \sim 1.4\text{--}5$ amplifies the cluster-edge magnetic fields ($\sim 0.1 \mu\text{G}$) to the levels required for the observed synchrotron luminosity ($B \sim 0.2\text{--}0.5 \mu\text{G}$). The bridge morphology follows from the overlap of the S^3 patches of the two clusters: in the overlapping region, both clusters contribute enhanced fields, producing a constructive interference pattern aligned along the inter-cluster axis.

3.5 Anomaly 5: The Cosmological Seed-Field Problem

Observed magnetic fields of $\sim \mu\text{G}$ strength in galaxies and galaxy clusters require seed fields to be amplified by dynamo action. However, the required seed strength ($\gtrsim 10^{-20}$ G) exceeds what most early-universe mechanisms (e.g., the Biermann battery, electroweak baryogenesis) can naturally produce [16, 17]. Recent work has suggested that primordial fields of 5–10 pG may suffice without dynamo amplification [18], but the generation of even these field strengths remains theoretically problematic.

DST interpretation. The S^3 enhancement relaxes the seed-field requirement. A weaker primordial seed B_0 generates an *observed* field $B_{\text{obs}} = B_0 \cdot \eta(r)^{3/2}$ when probed at cosmological distances. For $\eta^{3/2} \sim 10\text{--}100$, the required intrinsic seed drops to $10^{-22}\text{--}10^{-21}$ G, well within the range producible by standard electroweak mechanisms. The seed-field problem is not that seeds are too weak, but that observers have been using the wrong propagation law (flat $1/r^3$ instead of the S^3 -enhanced cotangent law).

4 Unified Interpretation: The S^3 Enhancement as a Single Geometric Origin

4.1 The Electromagnetic Analog of Flat Rotation Curves

The five anomalies surveyed in Section 3 share a common phenomenological signature: electromagnetic fields are observed to be stronger than the inverse-square (or inverse-cube, for dipoles) law predicts, by factors of $\sim 3\text{--}100$ depending on the

distance scale. This is the electromagnetic analog of the flat galactic rotation curve problem in gravity: in both cases, the long-range field is “too strong” relative to the Newtonian/Coulombic expectation.

In the double spacetime theory, both anomalies have the same geometric origin: the compact S^3 topology of the dual spacetime, which enhances all long-range interactions via the universal factor $\eta(r) = [r/(R \sin(r/R))]^2$. The cotangent potential replaces $1/r$ for all interactions coupled to the dual rotor, whether gravitational, electromagnetic, or otherwise.

Table 1: Gravitational and electromagnetic anomalies unified by the S^3 enhancement factor $\eta(r)$.

Phenomenon	Gravitational analog	EM observation
Excess at large r	Flat rotation curves	Void magnetic fields $\gtrsim 10^{-14}$ G
Isotropic excess	—	ARCADE 2 radio background ($\times 5\text{--}6$)
Non-evolution with z	—	Cluster B -fields constant to $z > 0.9$
Large-scale structure	Cosmic web	Mpc radio bridges
Seed problem	Missing mass (“dark matter”)	Seed-field problem

4.2 Constraining R_{EM} from Observations

The five anomalies probe different distance scales, providing independent constraints on R_{EM} :

- **Void magnetic fields** ($r \sim 1\text{--}10$ Mpc, $\eta^{3/2} \sim 10\text{--}100$): $R_{\text{EM}} \sim 1\text{--}5$ Mpc.
- **ARCADE 2** ($r \sim 100$ Mpc–1 Gpc, $\eta \sim 5\text{--}6$): $R_{\text{EM}} \sim 50\text{--}500$ Mpc.
- **Cluster B -field non-evolution** ($r \sim 1\text{--}3$ Gpc, $\eta^{3/2} \sim 3\text{--}5$): $R_{\text{EM}} \sim 1\text{--}2$ Gpc.
- **Radio bridges** ($r \sim 2\text{--}3$ Mpc, $\eta \sim 2\text{--}5$): $R_{\text{EM}} \sim 1\text{--}3$ Mpc.

These estimates span a wide range, suggesting that R_{EM} may depend on the source properties (total magnetic moment, spatial extent) analogously to the gravitational case where $R_{\text{grav}} = \sqrt{GM/a_0}$ depends on the total mass. A natural electromagnetic analog, $R_{\text{EM}} = \sqrt{\mu_0 M/a_{\text{EM}}}$, would yield source-dependent compactification radii consistent with the observed hierarchy: small R_{EM} for individual galaxies (Mpc scale, relevant for voids and bridges), large R_{EM} for the integrated cosmic source population (Gpc scale, relevant for ARCADE 2 and cluster evolution).

4.3 Azimuthal Equidistant Projection: The Geometric Picture

The enhancement factor $\eta(r)$ has a transparent geometric interpretation. Consider the azimuthal equidistant projection of S^2 centered on the North Pole: near the pole, the map faithfully represents distances and areas; at the South Pole (the antipodal point), the single point is mapped to the entire circumference of the projection boundary, producing infinite distortion.

In three dimensions, each electromagnetic source sits at the “pole” of its own S^3 chart. The electromagnetic field, computed in the flat tangent-space (log) coordinates, appears progressively stronger at larger distances because the S^3 geometry compresses the flux surface area:

$$\frac{A_{\text{flat}}(r)}{A_{S^3}(r)} = \frac{4\pi r^2}{4\pi R^2 \sin^2(r/R)} = \eta(r).$$

The universe is tiled by overlapping S^3 electromagnetic patches — one centered on each source — forming a cosmic atlas whose boundary zones (where η is largest) coincide with the cosmic voids and filaments where the strongest anomalies are observed.

5 Falsifiable Predictions

5.1 Prediction 1: Distance-Dependent Enhancement of Faraday Rotation Measures

If the S^3 enhancement applies to electromagnetic fields, Faraday rotation measures (RMs) of background sources should show a systematic excess at large impact parameters from foreground magnetized structures. Specifically, the RM profile of a magnetized galaxy cluster should deviate from the standard $1/r^2$ decline:

$$\text{RM}(r) \propto \int n_e(l) B_{\parallel}(l) dl \sim \text{RM}_{\text{flat}}(r) \cdot \eta(r)^{3/2}.$$

This predicts an upturn or flattening of the RM profile at projected separations $r \gtrsim R_{\text{EM}}$, testable with forthcoming Square Kilometre Array (SKA) RM grid observations.

5.2 Prediction 2: Helicity Dependence of the Radio Excess

In the double spacetime theory, the dual map $X \mapsto Xi$ is intrinsically chiral (Section 11 of [1]). Left-handed and right-handed circularly polarized electromagnetic waves couple differently to the dual rotor: right-handed polarization reinforces the torsional mismatch, while left-handed polarization opposes it. Consequently, the S^3 enhancement factor should differ between the two helicities:

$$\eta_+(r) \neq \eta_-(r).$$

This predicts a *net circular polarization* in the ARCADE 2 excess, absent in all conventional astrophysical models. Detection of a frequency-independent circular polarization fraction at the $\sim 10^{-3}$ level in the isotropic radio background would constitute strong evidence for the S^3 dual-spacetime origin.

5.3 Prediction 3: Reversal of Magnetic Field Enhancement Beyond $\pi R_{\text{EM}}/2$

The cotangent potential crosses zero at $r = \pi R_{\text{EM}}/2$, beyond which the torsion scalar J changes sign (repulsive regime). For the electromagnetic sector, this implies that at distances exceeding $\pi R_{\text{EM}}/2$ from a magnetized source, the effective magnetic field should *weaken more rapidly* than $1/r^3$, eventually reversing sign (field reversal). This prediction is testable via stacking analyses of RM profiles at very large impact parameters ($\gtrsim 5$ Mpc) from galaxy clusters, using the dense RM grids expected from SKA Phase 2.

5.4 Prediction 4: Correlation Between ARCADE 2 Excess and Large-Scale Structure

While the ARCADE 2 excess is isotropic to first approximation, the S^3 patch structure predicts subtle anisotropies correlated with the distribution of nearby magnetized structures. Specifically, the radio brightness temperature should be slightly higher in directions aligned with the local cosmic filament network (where η is enhanced by overlapping S^3 patches) and lower toward the centers of nearby voids (where the observer sits near the “pole” of the nearest galaxy’s S^3 patch, and $\eta \approx 1$). Cross-correlating high-resolution maps of the isotropic radio background with galaxy surveys (e.g., DESI, Euclid) could reveal this signature.

6 Discussion

6.1 Relation to the Gravitational Sector

The electromagnetic anomalies presented here are not independent of the gravitational predictions of the double spacetime theory. Both arise from the same S^3 compactness of the dual spacetime, and both are parameterized by the universal enhancement factor $\eta(r)$. The consistency check is powerful: the *same* geometric framework that eliminates dark matter must *simultaneously* explain the cosmic magnetic field anomalies. Any tuning that explains one sector but fails the other would falsify the theory.

6.2 Why Laboratory Tests Cannot Detect the Effect

The most precise laboratory tests of Coulomb’s law constrain deviations at the level $|q| < 10^{-16}$ [19], where q parameterizes the exponent in $F \propto r^{-(2+q)}$. These tests are conducted at scales of ~ 1 m, where $r/R_{\text{EM}} \sim 10^{-22}$ and $\eta \approx 1 + \mathcal{O}(10^{-44})$. The S^3 deviation is ~ 44 orders of magnitude below the current experimental sensitivity, explaining why the effect is invisible in the laboratory but dominant at astrophysical scales.

6.3 Implications for the Standard Model of Cosmology

If the S^3 enhancement of electromagnetic fields is confirmed, the implications extend far beyond cosmic magnetism:

- **Cosmic microwave background:** Thomson scattering cross-sections at recombination are unaffected (local process), but the propagation of CMB photons over cosmological distances is subject to η -enhanced Faraday rotation, potentially contributing to the observed B -mode polarization anomalies.
- **Hubble tension:** The S^3 enhancement modifies the luminosity-distance relation for standard candles, mimicking the effect of accelerated expansion. A portion of the apparent acceleration attributed to dark energy may in fact be an S^3 projection effect.
- **Baryon acoustic oscillations:** The sound horizon at recombination, calibrated assuming flat electromagnetic propagation, would require correction if the S^3 enhancement applies to photon-baryon coupling at large scales.

These implications deserve dedicated investigation and are beyond the scope of the present paper.

7 Conclusions

The double spacetime theory, grounded in the biquaternion algebra $\text{Cl}(3, 1)$, predicts that all long-range interactions — not only gravity — are enhanced at large distances by the compact S^3 topology of the particle-intrinsic dual spacetime. The universal enhancement factor $\eta(r) = [r/(R \sin(r/R))]^2$ replaces the flat-space inverse-square law with the cotangent law on S^3 .

We have demonstrated that five major classes of astrophysical observations — void magnetic fields, the ARCADE 2 radio excess, non-evolving cluster magnetic fields, megaparsec radio bridges, and the seed-field problem — are all consistent with this single geometric mechanism. Each anomaly, previously requiring its own ad hoc explanation (primordial magnetogenesis, dark matter decay, turbulent dynamos, etc.), is now unified as a manifestation of the same S^3 compactness that eliminates dark matter from galactic dynamics.

The theory makes specific, falsifiable predictions: distance-dependent Faraday rotation enhancement, helicity-dependent radio excess, magnetic field reversal beyond $\pi R_{\text{EM}}/2$, and anisotropy in the radio background correlated with large-scale structure. Forthcoming observations with the SKA, LOFAR 2.0, and next-generation CMB experiments are ideally positioned to test these predictions within this decade.

If confirmed, the S^3 enhancement of cosmic electromagnetic fields would elevate the double spacetime theory from a gravitational reformulation to a universal framework governing all long-range physics, and would mark the end of the flat-space paradigm that has dominated astrophysics since Newton.

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